

Technical Architecture, Software Logic, and Patent Analysis of the 1996 Bandai Tamagotchi (Gen 1 / Gen 2)

Game Logic, State Machines & Flowcharts

Hardware Architecture and Microcontroller Constraints

The original 1996 Bandai Tamagotchi (Generation 1 and Generation 2) was built upon an ultra-low-power, resource-constrained embedded computing platform designed to balance operational longevity with deterministic state execution¹. At the core of the system sits a 4-bit CMOS single-chip microcontroller from the Seiko Epson E0C600 series, specifically the Seiko Epson E0C6S46³. The microcontroller is driven by an external 32.768 kHz tuning-fork crystal oscillator, which provides the master clock reference for both the central processing unit and the integrated real-time clock (RTC) hardware³.

The system memory layout is tightly constrained by its 4-bit architecture³. Program memory consists of a mask-programmed ROM containing 6,144 words, where each instruction word is 12 bits wide (equivalent to 7.3728 KB)³. Internal data storage relies on a static RAM array of 640 4-bit nibbles (320 bytes total), which holds all active state registers, counters, care variables, and flag bits³. A separate, dedicated Display RAM block of 160 4-bit nibbles (80 bytes) serves as a direct-mapped frame buffer for the visual output⁵. The display hardware features a custom monochrome Liquid Crystal Display (LCD) with a central 32×16 dot-matrix pixel grid surrounded by eight static peripheral icon indicators (four positioned along the top margin and four along the bottom margin)⁴.

User interaction is facilitated through three primary tactile momentary micro-switches—designated key switches SW1, SW2, and SW3—along with a recessed hardware reset switch (SW4) wired directly to the CPU's master reset line⁷. Audio synthesis is handled by a single-channel piezoelectric buzzer (BZ1) driven by square-wave pulse generators integrated directly into the microcontroller⁴. Power is supplied by two 1.5V alkaline button cells (LR44), requiring the software to execute aggressive power-saving protocols during idle states⁷.

Hardware Subsystem	Architectural Specification	Technical Function in System Architecture
Central Processor	Seiko Epson E0C6S46 (4-bit CMOS)	Executes mask ROM instructions, manages state transitions, and processes

	Microcontroller) ³	I/O ³
Master Clock	32.768 kHz Crystal Oscillator ³	Drives instruction timing, timer interrupts, and the Real-Time Clock counter ⁷
Program ROM	6,144 words × 12 bits (7.3728 KB) ³	Stores application code, character sprite sheets, and audio frequency tables ⁵
General Data RAM	640 nibbles × 4 bits (320 bytes) ³	Holds runtime variables, lifecycle timers, decay counters, and status flags ³
Display Buffer RAM	160 nibbles × 4 bits (80 bytes) ⁵	Hardware-mapped frame buffer driving the 32×16 dot-matrix display grid ⁴
Liquid Crystal Display	32×16 Dot Matrix + 8 Static Icons ⁴	Visual rendering unit for character animations and menu selection states ⁷
Input Switches	3 Momentary Switches (SW1, SW2, SW3) + Reset (SW4) ⁷	Hardware interrupt sources for user menu navigation, input confirmation, and system reset ⁷
Audio Output	Piezoelectric Buzzer Transducer (BZ1) ⁴	Generates audio feedback, call chimes, and mini-game tone sequences ¹

Software Main Execution Loop and Real-Time Logic

The operational core of the Tamagotchi software is an interrupt-driven real-time execution loop⁷. Because the E0C6S46 lacks a hardware operating system, execution relies on deterministic state polling synchronized with hardware timer ticks generated by the 32.768 kHz oscillator³. A hardware prescaler divides the crystal frequency down to a 1 Hz interrupt signal, which increments the internal Real-Time Clock (RTC) registers stored in RAM (seconds, minutes, hours, and total age in days)⁷.

The software execution sequence operates continuously through a six-stage loop:

First, the Real-Time Clock update routine fires upon every 1 Hz interrupt tick⁷. The CPU increments time counters and checks for minute and hour register overflows⁷. These transitions

trigger scheduled background checks for variable decay, sleep schedules, feces generation, and discipline calls⁷.

Second, the system executes the state variable decay evaluation routine⁷. The CPU fetches the active character profile ID and retrieves its specific decay parameters from ROM lookup tables⁷. Internal timers tracking hunger, happiness, discipline, and health are evaluated against current tick counts¹². If a variable threshold is reached, the corresponding variable in RAM is decremented or updated⁷.

Third, the CPU polls the I/O ports mapped to key switches SW1, SW2, and SW3⁷. Button state transitions undergo software debouncing delays⁸. If a valid button press is registered, the software interrupts the current idle animation and routes control to the context-sensitive input handler⁸.

Fourth, the behavioral logic and event dispatch engine evaluates whether an attention alert must be generated⁷. If hunger or happiness reaches zero, or if a misbehavior call is triggered, the system sets the internal attention flag, illuminates the peripheral Call Icon, and invokes the audio driver to play a beep sequence on the piezoelectric buzzer¹.

Fifth, the display rendering pipeline calculates the next frame of animation⁵. The system combines character base sprites from ROM with environmental status overlays (such as accumulated feces, sickness skulls, or speech bubbles) into the 160-nibble display RAM buffer, which automatically updates the LCD⁵.

Sixth, if no user input or active alert is pending, the CPU executes a low-power HALT instruction⁹. The processor remains in a dormant state until the next 1 Hz timer interrupt or button line interrupt brings the CPU back into full operational mode, minimizing power consumption⁷.

Care Variable Mechanics and Decay Algorithms

The behavioral simulation relies on six primary care variables stored as 4-bit nibbles in RAM³. Because memory registers are constrained to 4-bit values, variables are bounded within strict

integer ranges (0 to 15, or $0x0$ to $0xF$)³:

The Hunger Register is an integer value ranging from 0 to 4, rendered on screen as four heart icons¹². Feeding the pet a Meal restores 1 hunger heart and adds 1 gram of weight¹².

The Happiness Register is an integer value ranging from 0 to 4, rendered as four heart icons¹². Feeding a Snack restores 1 happiness heart but adds 2 grams of weight¹². Successfully playing a mini-game restores 1 happiness heart and reduces weight by 1 gram¹².

The Discipline Register is stored as an integer ranging from 0 to 4, corresponding to visual meter fill levels of 0%, 25%, 50%, 75%, and 100%¹². Administering a Scold during a misbehavior event increases this value by 1 unit (25%)¹².

The Weight Register tracks body mass in grams¹². Minimum weight boundaries are enforced based on the current evolutionary stage, while the absolute global maximum mass is capped at 99 grams¹².

The Feces and Health Flags consist of a poop counter (tracking 0 to 4 uncleaned piles on screen) and a binary sickness flag (0 = Healthy, 1 = Sick)⁷.

The Sleep and Illumination Register tracks whether the character is currently in a sleep state and whether the screen lights have been switched off (0 = Lights On, 1 = Lights Off)⁷.

The depletion rate of hunger and happiness depends heavily on the active evolutionary stage¹². During early stages, decay occurs rapidly to force frequent player interaction, whereas adult forms feature decay rates that reflect the quality of the care provided during development¹².

Evolutionary Stage	Hunger Heart Decay Interval	Happiness Heart Decay Interval	Minimum Weight	Scheduled Sleep Window
Stage 0: Egg	Infinite (Static State) ⁷	Infinite (Static State) ⁷	N/A	Active 24/7 (No Sleep) ⁷
Stage 1: Baby	3 to 5 Minutes ⁷	3 to 5 Minutes ⁷	5 grams	8:00 PM – 9:00 AM
Stage 2: Child	15 to 20 Minutes	15 to 20 Minutes	10 grams	8:00 PM – 8:00 AM
Stage 3: Teenager	30 to 45 Minutes	30 to 45 Minutes	20 grams	9:00 PM – 9:00 AM
Stage 4: Adult (High Tier: Mametchi / Mimitchi)	60 Minutes	60 Minutes	30 grams	10:00 PM – 9:00 AM ¹²
Stage 4: Adult (Mid Tier: Ginjirotchi / Pochitchi)	45 Minutes	45 Minutes	30 grams	10:00 PM – 9:00 AM ¹²
Stage 4: Adult (Low Tier: Maskutchi / Zukitchi)	30 Minutes	30 Minutes	20 grams	11:00 PM – 11:00 AM ¹²

Stage 4: Adult (Poor Tier: Nyorotchi / Kusatchi)	15 to 20 Minutes	15 to 20 Minutes	10 grams	10:00 PM – 9:00 AM
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Care Mistakes and Discipline Call Evaluation

Care mistakes and discipline calls directly drive the branching logic of the character's evolution¹².

A Care Call is initiated whenever either the Hunger or Happiness register drops to zero¹². The system illuminates the peripheral Call Icon, plays an audio alert, and initializes an internal 15-minute countdown timer¹. If the player responds by feeding or playing with the pet before the 15 minutes elapse, the Call Icon turns off and the timer resets¹². If the timer expires without player intervention, the Call Icon turns off automatically, and the internal RAM register tracking total Care Mistakes (Care_Mistake_Count) increments by one¹².

A Misbehavior Call is triggered at random intervals generated by pseudo-random seeds in ROM⁷. The pet sounds an alarm and turns on the Call Icon despite having positive Hunger and Happiness values, or refuses to eat when not full¹². The system opens a 15-minute response window¹². If the player selects the Discipline icon (scold) within this window, the Discipline meter increases by 25%, the call ends, and no mistake is recorded¹². If the 15-minute window expires without discipline, the opportunity is lost, and a Discipline Mistake is logged¹².

Feces generation occurs on fixed timer schedules following meals¹². Uncleaned feces accumulate up to a maximum of 4 items on screen¹². If 4 feces items accumulate, or if a single feces item remains uncleaned past a threshold timer, an illness evaluation routine sets the Is_Sick flag to true⁷. When sick, the pet displays a skull icon and rejects standard care inputs until cured with Medicine⁷. Allowing a pet to remain sick for extended periods causes rapid lifespan reduction and leads directly to death¹.

Evolutionary Branching Matrix and Lifecycle Transitions

The lifespan of the classic Tamagotchi follows a deterministic state machine progressing through six distinct evolutionary phases: Egg, Baby, Child, Teenager, Adult, and Death or Departure⁷. Branching decisions evaluated at stage transitions depend on the cumulative Care Mistakes and Discipline percentages accumulated in RAM during preceding stages¹².

Generation 1 (P1) Evolutionary Branching Logic

Development begins in the Egg state, which displays a static egg graphic for 5 minutes post-time setup before hatching into the Baby character, Babytchi⁷. The Baby stage lasts exactly 60 minutes and requires intensive care⁷. Care mistakes made during this hour set base health parameters but do not alter the transition into the Child character, Marutchi⁷.

The Child stage lasts between 24 and 48 hours¹². Upon reaching the end of the Child timer, the

system evaluates Care_Mistake_Count:

- If Care Mistakes are less than or equal to 2, the pet evolves into the high-care Teenager, Tamatchi¹².
- If Care Mistakes are greater than or equal to 3, the pet evolves into the low-care Teenager, Kuchitamatchi¹².

The Teenager stage lasts between 48 and 72 hours¹². The final transition to the Adult stage evaluates the total accumulated Care Mistakes and the final Discipline meter fill level¹²:

Transitions from Tamatchi (High-Care Teen):

- **Mametchi**: Requires 0 Care Mistakes and a Discipline meter of 75% to 100%¹².
- **Ginjirotchi**: Requires 0 to 1 Care Mistakes and a Discipline meter below 75%¹².
- **Maskutchi**: Requires 2 to 3 Care Mistakes¹².

Transitions from Kuchitamatchi (Low-Care Teen):

- **Kuchipatchi**: Requires 0 to 1 Care Mistakes and a Discipline meter of 75% to 100%¹².
- **Nyorotchi**: Requires 2 to 3 Care Mistakes and a Discipline meter below 75%.
- **Tarakotchi**: Requires 4 or more Care Mistakes.

Special Secret Character Transition:

- **Bill**: If Maskutchi is cared for impeccably and its Discipline meter is brought to 100% before its lifespan timer expires, it evolves into the hidden character, Bill¹².

Generation 2 (P2) Evolutionary Branching Logic

Generation 2 follows identical timing structures but replaces character graphics and adjusts branching thresholds¹⁶. The Baby form (Tonmarutchi) evolves into the Child form (Tongaritchi)¹⁷.

Transitions from Tongaritchi (High-Care Child/Teen Tier):

- **Mimitchi**: Perfect care tier (0 Care Mistakes, 100% Discipline), equivalent to Mametchi.
- **Pochitchi**: High care tier (0–1 Care Mistakes, $< 75\%$ Discipline)¹⁵.
- **Zukitchi**: Moderate care tier (2–3 Care Mistakes)¹⁷.

Transitions from Hashitamatchi (Low-Care Child/Teen Tier):

- **Hashizoutchi**: High discipline tier from low-care teen¹⁷.
- **Kusatchi**: Moderate care tier from low-care teen¹⁷.
- **Takotchi**: Low care tier (≥ 4 Care Mistakes)¹⁷.

Special Secret Character Transition:

- **Sekitori**: If Zukitchi achieves 100% Discipline before the end of its adult lifecycle, it transforms into the secret character, Sekitori¹⁷.

Official Patents & Design Documents

International Patent Registration and Legal Documentation

To prevent market infringement and establish intellectual property rights over the virtual pet simulation mechanics, Bandai Co., Ltd. and WiZ Co., Ltd. filed global patent applications

between 1996 and 1997¹. The primary intellectual property was developed by Akihiro Yokoi (founder of WiZ) and Aki Maita (concept developer at Bandai)¹.

Patent Number	Filing / Grant Dates	Designated Inventors	Assignee Entities	Core Technical Claims & Legal Scope	Database Links
US5966526 A	Filed: Jun 11, 1997 Granted: Oct 12, 1999 ⁷	Akihiro Yokoi ⁷	Bandai Co., Ltd., WiZ Co., Ltd. ⁷	System architecture , main control loop, discipline evaluation routines, care-respon se timing windows, and physical two-wire peer coupling interface ⁷ .	Google Patents US5966526 A
US6213871 B1	Filed: Feb 19, 1997 Granted: Apr 10, 2001 ⁸	Akihiro Yokoi ⁸	Bandai Co., Ltd. ⁸	State transition evaluation algorithms, dynamic memory register mapping, mark-displa y menu interaction, and multi-stage growth branching	Google Patents US6213871 B1

				matrices ⁸ .	
JP1997005357U	Filed: Jun 6, 1997 ⁷	Akihiro Yokoi, Aki Maita ¹	Bandai Co., Ltd. ⁷	Priority Japanese utility model protecting hardware layout, PCB button geometry, and LCD casing design ⁷ .	Espacenet Patent Search
USD398659S	Filed: Mar 21, 1997 Granted: Sep 22, 1998 ³	Akihiro Yokoi, Aki Maita ¹	Bandai Co., Ltd. ⁷	Design patent protecting the ornamental egg-shaped housing geometry and three-button front panel configuration ¹ .	Google Patents USD398659S

Analysis of Patent Schematics and Circuit Diagrams

The circuit schematics detailed in patent filings US5966526A and US6213871B1 outline the electronic hardware implementation required to run the simulation engine⁷.

Electrical Circuit Architecture (Ref: FIG. 5 of US5966526A / FIG. 4 of US6213871B1)

The system relies on a unified integrated microcomputer (21) housing the core processing units⁷:

The Control Microcomputer Unit (21) houses the Central Processing Unit (5), internal program ROM, data memory (3), real-time clock counter circuits, and LCD driver circuitry on a single silicon die⁷.

The Clock Pulse Generator (XL) consists of an external 32.768 kHz quartz crystal resonator connected across microcomputer terminals to drive system processing and time tracking⁷.

The Power Distribution System (BT) utilizes two direct-current button cells filtered by smoothing capacitors (CC1 through CC12) to ensure stable logic levels during piezo audio discharge⁷. The Control Switches (SW1, SW2, SW3) connect directly to microcomputer input ports with internal pull-up resistors⁷. The hard reset switch (SW4) connects directly to the master reset pin to re-initialize memory⁷.

The Audio Transducer Circuitry consists of a piezoelectric buzzer (BZ1) connected through current-limiting resistors (RR4, RR5) to dedicated sound-generator output pins⁷.

The Physical Peer-to-Peer Interface (Connecting Means 7), detailed in US5966526A, features a two-wire serial communication port with a signal line (7a) and ground line (7b) tied to CPU I/O terminals via isolation diodes (DD1, DD2) and resistor RR6⁷. This hardware connection allows two units to exchange character data registers for breeding and coupling operations⁷.

Software Flowchart Control Logic (Ref: FIG. 6 through FIG. 10 of US5966526A)

The software structure documented in the patent filings establishes a systematic state control process:

In Step SP1, the CPU continuously monitors hardware reset switch SW4⁷. When activated, the CPU clears RAM registers and branches to the setup initialization state⁷.

In Step SP3, the system invokes time-setting mode⁷. The player manipulates key switches SW1 (hours) and SW2 (minutes), confirming with SW3 to store initial values in the RTC registers⁷.

In Step SP5, the display unit renders the initial egg graphic on the LCD matrix⁷.

In Step SP7, the egg hatches into newborn character KT1 (A1), initiating the First Growth Procedure⁷. Active decay timers begin tracking hunger and happiness depletion, while user care actions (feeding, cleaning, medical treatment) are processed through input polling loops⁷.

In Step SP9, upon elapsed time reaching approximately 60 minutes, character KT1 evolves into Child character B1⁷.

In Step SP11, the system evaluates historical care metrics (response times, discipline applications, care mistakes) logged during earlier stages⁷. The engine selects an adult profile from the character matrix in ROM and loads the corresponding adult lifecycle routine⁷.

Design Specifications and Input Interface Mechanics

The user interface was engineered to enable full navigation of a multi-variable simulation using only three physical buttons¹.

LCD Display Structure and Peripheral Icon Layout

The visual display consists of a central 32×16 dot-matrix display surrounded by eight peripheral icon indicators split into two horizontal rows⁷:

Top Peripheral Icon Row:

- **Icon 1 (Status Check - Mark 17a):** Opens status screens displaying age, weight, discipline meter, hunger hearts, and happiness hearts⁷.
- **Icon 2 (Food Selection - Mark 17b):** Opens sub-menu to choose between a Meal (restores hunger) or a Snack (restores happiness)⁷.

- **Icon 3 (Game Mode - Mark 17c):** Initiates the directional guessing mini-game to restore happiness and reduce weight⁷.
- **Icon 4 (Restroom Clean - Mark 17d):** Triggers an animation that sweeps accumulated feces off the screen⁷.

Bottom Peripheral Icon Row:

- **Icon 5 (Discipline / Scold - Mark 17e):** Administers scolding to increase the discipline meter when the pet calls improperly⁷.
- **Icon 6 (Medicine - Mark 19a):** Administers medical injections to cure active sickness⁷.
- **Icon 7 (Illumination - Mark 19b):** Toggles the screen light ON or OFF when the pet enters its sleep state⁷.
- **Icon 8 (Attention Alert - Mark 19e):** Automatically lit by the system when care intervention is required⁷.

Input Switch Functional Logic

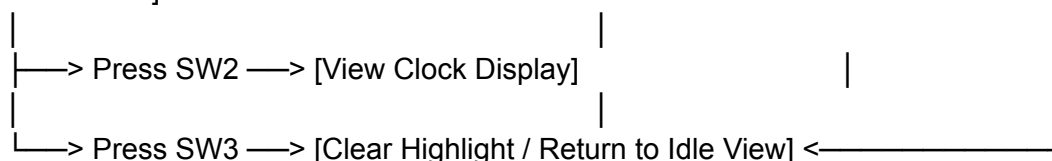
The three input switches (SW1, SW2, SW3) govern navigation across all system states⁷:

Switch SW1 (Button A) functions as the Select input⁸. Pressing Button A cycles highlighting sequentially through peripheral icons 1 through 7⁸. Within sub-menus, Button A moves the cursor between options¹². In time-setting mode, Button A increments hours⁷.

Switch SW2 (Button B) functions as the Execute input⁸. Pressing Button B activates the currently highlighted peripheral icon⁸. If no icon is highlighted, pressing Button B directly opens the Real-Time Clock view⁷. In time-setting mode, Button B increments minutes⁷.

Switch SW3 (Button C) functions as the Cancel input⁸. Pressing Button C deselects active icons, closes sub-menus, and returns the display to the main character idle view⁸. Pressing Button A and Button C simultaneously toggles system audio muting ON or OFF.

[Main Idle View] —> Press SW1 —> [Cycle Icons 1-7] —> Press SW2 —> [Execute Selected Action]



This interaction loop—combining simple physical inputs with complex care decay algorithms and patented multi-tier growth logic—enabled the classic Tamagotchi to operate an engaging virtual life simulation within a low-cost, low-power embedded platform¹.

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